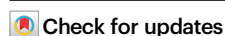


Unveiling the Brønsted acid mechanism for Meerwein–Ponndorf–Verley reduction in methanol conversion over ZSM-5

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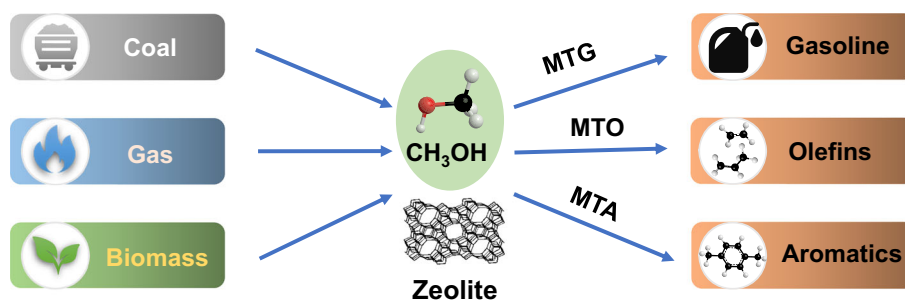
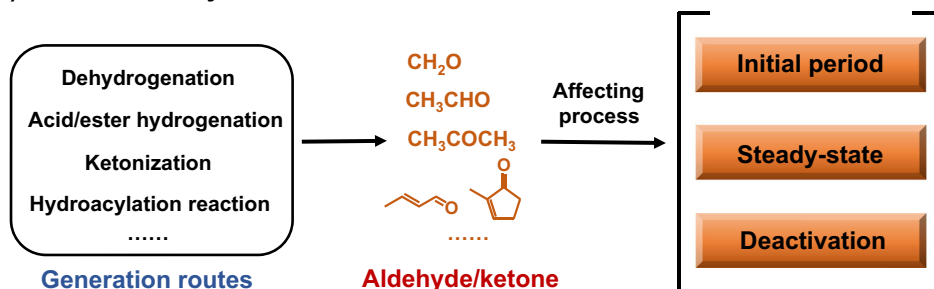
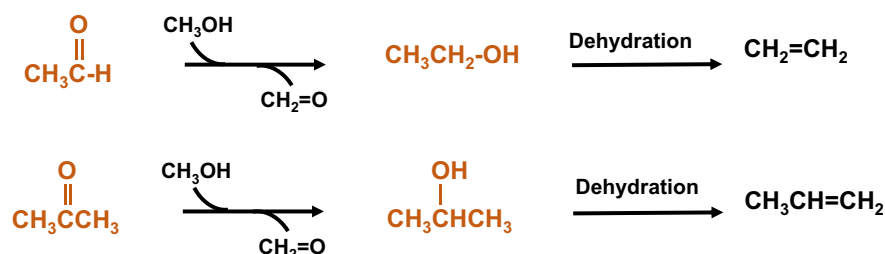
The conversion of methanol over zeolites offers a sustainable alternative for fuels and chemicals production. However, a complete understanding of the competing reaction pathways, particularly those leading to C-C bond formation, remains elusive. This work presents a novel mechanism for selective methanol conversion in ZSM-5 zeolites, involving a Brønsted acid site (BAS)-mediated Meerwein–Ponndorf–Verley (MPV) reduction pathway. Employing a multidimensional solid-state NMR spectroscopy combined with isotopic labeling and theoretical calculations, we identify this pathway for acet-aldehyde reduction with methanol, directly contributing to ethene formation. This mechanism, involving carbenium ion intermediates like 1-hydroxyethane or 1-methoxyethane ions, contrasts with the well-established Lewis acid-catalyzed MPV process. Based on reactant adsorption modes, we propose two distinct reaction routes for BAS-MPV reduction, bridging the gap between direct and hydrocarbon pool mechanisms in methanol conversion. We further demonstrate the applicability of this pathway to acetone, highlighting its broader role in the early stages of the reaction.

With the increasing global energy demand and the dwindling supplies of conventional petroleum resources, it becomes imperative to explore alternative carbon-based feedstocks that can effectively meet the world's growing needs for fuels and chemicals. Conversion of methanol over zeolites has attracted much attention, which can be flexibly tailored to yield light olefins (Methanol-to-Olefin, MTO), gasoline (Methanol-to-Gasoline, MTG), or aromatics (Methanol-to-Aromatics, MTA) (Fig. 1a)^{1–4}. Notably, the methanol can be derived from a wide range of resources, including coal, natural gas, and biomass. However, despite zeolites' high activity in methanol conversion, challenges arise in terms of product selectivity loss and rapid coking over zeolites^{5–7}. Understanding the reaction mechanism of methanol conversion is essential for the development of high-performance catalysts and optimization of chemical process, but is rather challenging due to the complex reaction network involved in such reaction^{3,8–10}.

The methanol conversion is a well-established example of autocatalysis^{11,12}. Two primary mechanisms govern this process: the direct mechanism^{12–21} and the indirect mechanism, also known as the hydrocarbon pool (HP) mechanism^{22–30}. The direct mechanism is believed to be responsible for the initial formation of C-C bonds. Over 20 potential reaction pathways have been proposed, each involving the coupling of C1 intermediates adsorbed on the zeolite catalyst surface (e.g., surface methoxy species, CO, formaldehyde)^{8,18,31–35}. These newly formed C-C bond species subsequently occupy active sites and participate in the conversion of methanol/DME to final products. In contrast, the indirect mechanism involves the participation of pre-existing C-C bond species, such as alkenes, aromatics, and carbocations, which act as catalytic platforms^{36–39}. C1 reactants (methanol/DME) are incorporated into these platforms and subsequently cleaved to yield final products. A dual-cycle model based on alkenes and

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(a) Methanol conversion over zeolite**(b) Roles of aldehyde/ketone in methanol conversion****(c) MPV reduction in methanol conversion (investigated in this work)****Fig. 1 | Roles of aldehyde/ketone in methanol conversion over zeolites.**

a Methanol conversion process to produce fundamental chemicals. MTG, MTO, and MTA mean conversion of methanol to gasoline, olefins, and aromatics

respectively. **b** Sources of aldehyde/ketone as well as their roles in methanol conversion. **c** MPV reduction for the transformation of aldehyde/ketone under a methanol atmosphere.

aromatics has been proposed, where the alkene-based cycle favors the formation of heavier products (C_3^+), while the aromatic-based cycle primarily produces ethene^{28–30}. Recent studies have highlighted the role of oxygenate pathways in the methanol conversion^{13,14,16,17,40–42}. These pathways are crucial for the formation of initial C-C bonds, hydrocarbon pools, and coke precursors. Common oxygenates involved in the reaction include acids, aldehydes/ketones, and their methylated derivatives (esters, acetals/ketals). Despite their lower abundance, oxygenates significantly impact product selectivity and catalyst deactivation.

Methanol, the primary reactant in methanol conversion, is the main source of oxygen found in the oxygenate intermediates. During the induction period, for example, methanol decomposition generates formaldehyde and CO, subsequently participate in the formation of C-C bond-containing oxygenates such as acetaldehyde, 1,2-dimethoxyethane, methyl acetate, and acetate^{13,14,16,41}. Elucidating the fate of these oxygenates during methanol conversion is paramount to achieving a comprehensive understanding of methanol conversion chemistry. Mechanistic studies demonstrated the transformation of methyl acetate/acetate into hydrocarbons via methylation and decarbonylation pathways^{43,44}. Ketene has been recognized as a potential key intermediate in this process^{15,45}. The recent application of

operando photoelectron photoion coincidence (PEPICO) spectroscopy provided definitive evidence for the presence of ketene within the reaction system, further solidifying its importance in methanol conversion¹⁷.

Zeolites employed as catalysts in methanol conversion promote the formation of another significant class of oxygenate intermediates: aldehydes/ketones and acetals/ketals (Fig. 1b). These molecules can be generated through various pathways, including acid/ester hydrogenation, carboxylic acid ketonization, and the hydroacylation reaction between alkenes and formaldehyde^{15,40,41,46,47}. Acetaldehyde, along with its methylated counterparts, is widely recognized as a primary intermediate responsible for the initial generation of hydrocarbons in methanol conversion^{13,17,41}. This concept is exemplified by studies using IR spectroscopy over SAPO-34, where 1,2-dimethoxyethane was identified as the first species containing a C-C bond. This intermediate facilitates the formation of primary propene¹³. Similarly, the hydrogenation of methyl acetate/acetate over ZSM-5 leads to the production of acetaldehyde, which plays a pivotal role in the formation of initial hydrocarbons⁴⁶. Recent research also suggests that ketonization of methyl acetate or acetic acid over ZSM-5 can lead to the formation of acetone, which can subsequently transform into alkenes and aromatics⁴⁸. The hydroacylation reaction offers an additional pathway

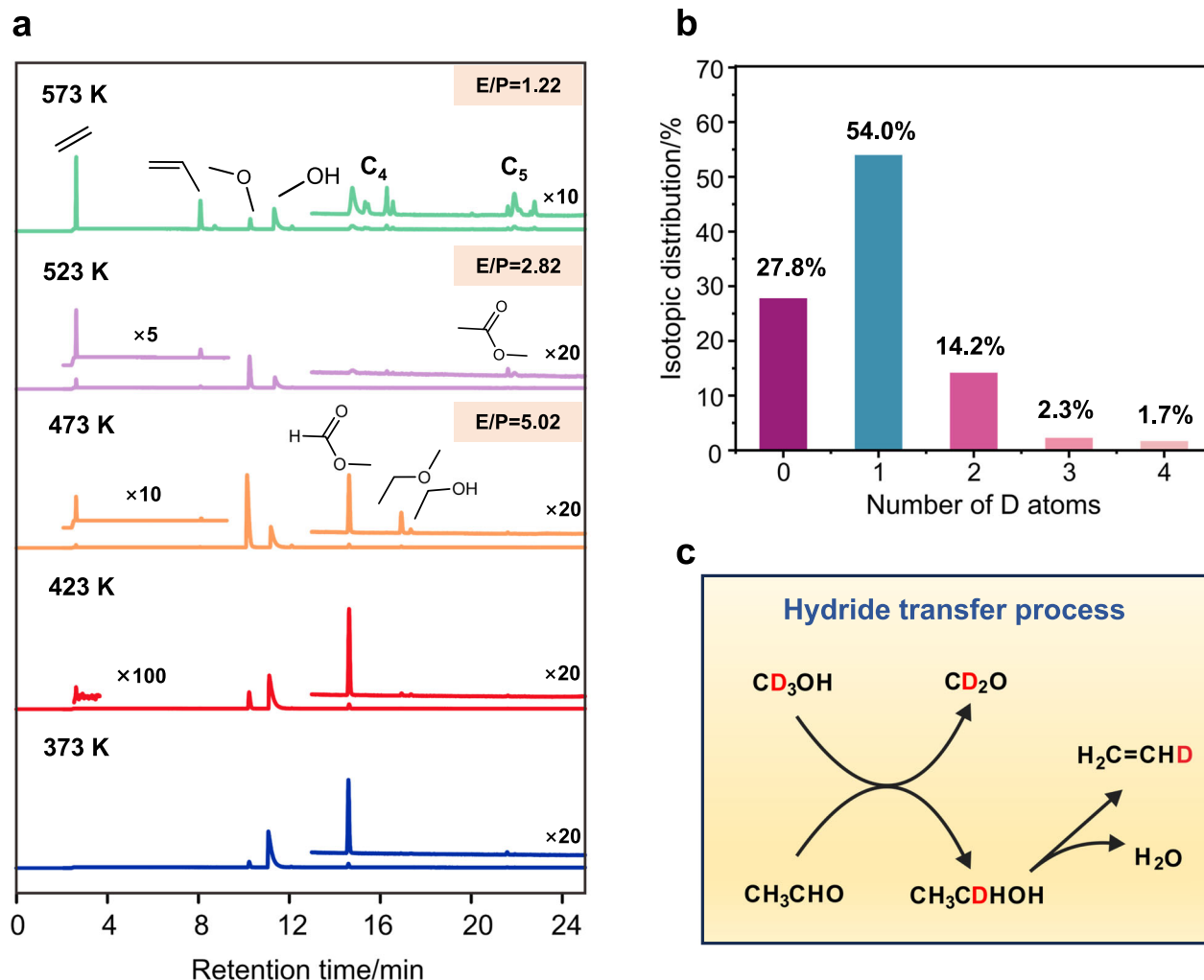


Fig. 2 | MPV reduction between acetaldehyde and methanol over ZSM-5. a GC analysis of the effluent products obtained from co-reaction of methanol and acetaldehyde (v/v, 4/1) over ZSM-5 at various temperatures for 8 s. E/P indicates the

ethene to propene ratio. **b** Deuterium atom distribution in ethene produced from co-reaction of CD_3OH with natural abundance acetaldehyde (v/v, 4:1) over ZSM-5 at 523 K for 8 s. **c** Illustration of the hydride transfer process for the MPV reaction.

for the formation of aromatic precursors. In this reaction, formaldehyde reacts with alkenes to yield higher aldehyde, which is favorable for the formation of aromatics during methanol conversion over zeolites^{40,47,49}. Although numerous studies have documented the high reactivity of aldehydes and ketones on zeolites^{13,41,46,50}, their precise mechanistic contribution to the overall methanol conversion pathway remains elusive. Notably, a crucial knowledge gap exists regarding the interactions between these oxygenates and methanol during the methanol conversion, particularly during the initial stages where methanol/DME are abundantly present. A deeper understanding of these interactions is essential for fully elucidating the complete contribution of aldehydes and ketones to hydrocarbon formation in methanol conversion over zeolites.

This work explores a previously unexplored facet of methanol conversion chemistry. We propose a novel Brønsted acid site (BAS)-mediated Meerwein-Ponndorf-Verley (MPV) reduction pathway within ZSM-5 zeolites, facilitating the conversion of acetaldehyde to ethene (Fig. 1c). This mechanism, distinct from the traditional Lewis acid-catalyzed MPV process, is elucidated through a combination of multi-dimensional solid-state NMR spectroscopy, isotopic labeling, and theoretical calculations. Furthermore, we demonstrate the applicability of this BAS-MPV pathway to acetone conversion, suggesting its broader role in the early stages of methanol transformation over acidic zeolites.

Results

BAS catalyzed MPV reduction over ZSM-5 zeolite

Proton-type ZSM-5 zeolites (Si/Al = 11.5) were utilized as catalysts to investigate the behavior of acetaldehyde during the methanol conversion. Detailed structural and acidity analyses are provided in Table S1 and Figs. S1–S3. The ^{27}Al MAS NMR spectra indicate the presence of only tetra-coordinated Al species at 55 ppm in the ZSM-5 zeolite (Fig. S2). The concentrations of BAS ($590 \mu\text{mol}\cdot\text{g}^{-1}$) and Lewis acid sites (LAS) ($6 \mu\text{mol}\cdot\text{g}^{-1}$) were determined using IR spectroscopy of adsorbed pyridine. The high concentration of BAS and negligible LAS suggest that the methanol reaction would be predominantly catalyzed by the BAS on ZSM-5 (Table S1). Methanol-only (Fig. S4) and methanol-acetaldehyde ($V_{\text{methanol}}/V_{\text{acetaldehyde}} = 4:1$) co-feeding (Fig. 2) experiments were conducted over ZSM-5 at elevated temperatures of 373, 423, 473, 523, and 573 K. To minimize secondary reactions and capture intermediates, the reactions were performed for a short duration of 8 s using a pulse-quench reactor, where the catalytic bed was rapidly quenched with liquid nitrogen within 0.5 s. The reactor effluent products were analyzed using connected gas chromatography (GC).

Methanol alone did not yield hydrocarbons, aside from dimethyl ether (CH_3OCH_3), at temperatures below 573 K (Fig. S4), indicating that an active HP mechanism is absent under these conditions. However, introducing acetaldehyde significantly impacts methanol conversion over the zeolites. As shown in Fig. 2a, substantial amounts of methyl

formate (HCOOCH_3) form during the co-reaction even at 373 K. This indicates the oxidation of methanol by acetaldehyde, potentially through a two-step pathway involving initial conversion to formaldehyde (confirmed by GC analysis, Fig. S5). Subsequent oxidation generates formate, which then reacts with methanol to form methyl formate. Concurrently, acetaldehyde is reduced to ethanol or surface ethoxy species on the zeolites, as confirmed by subsequent ^{13}C NMR analysis of the trapped species on the samples. This reveals a hydride transfer (i.e., MPV) reaction between acetaldehyde and methanol, a fundamental transformation in organic chemistry. While MPV reactions have frequently been reported over Lewis acid zeolites^{31,52}, this result demonstrates that BAS can also catalyze the MPV reaction in methanol conversion.

As the temperature rises to 423 K, methyl formate yield increases, reflecting enhanced methanol oxidation. Ethanol production also becomes evident. Further dehydration of ethanol with methanol leads to ethyl methyl ether ($\text{CH}_3\text{CH}_2\text{OCH}_3$) and ethene. Notably, ethene dominates the hydrocarbon products with an ethylene to propylene (E/P) ratio of 5.02 at 473 K, suggesting the MPV pathway as the primary route for hydrocarbon production. This ethene selectivity persists at higher temperatures, indicating continued dominance of the MPV mechanism. However, propene and heavier hydrocarbons also increase with temperature, leading to a declining E/P ratio (from 5.02 to 1.22). This suggests the concurrent participation of the HP mechanism at higher temperatures, responsible for C_3+ hydrocarbon production and the decrease in ethene selectivity^{30,53}. To further investigate the relative activity of BAS and LAS in the MPV reaction, we conducted a comparative study. The parent ZSM-5 zeolite was subjected to calcination at 973 K. This high-temperature calcination process caused the decomposition of the BAS, leading to the formation of LAS on the ZSM-5 (Table S1). To assess the activity of these different acid sites, we performed co-reactions between methanol and acetaldehyde using these ZSM-5 catalysts. The results demonstrate that the high-temperature calcined sample enriched with LAS, produce much less ethene compared to the parent zeolite (Fig. S6). This observation strongly suggests that the BAS is more active than LAS in the MPV reaction.

The observed MPV reduction suggests that acetaldehyde formed during methanol conversion reacts with unreacted methanol to produce ethene as the primary product. We hypothesize that during the initial stages of methanol conversion, when methanol is abundant, acetaldehyde preferentially interacts with surrounding methanol molecules, favoring the MPV reaction. To verify this, we conducted co-feeding experiments with a low acetaldehyde concentration (v/v, 1:100). As shown in Figure S7, ethene is virtually the sole alkene product, confirming the dominance of the MPV reaction under high methanol environments, resembling the early stages of methanol conversion.

To validate the hydride transfer mechanism between acetaldehyde and methanol, we employed isotopic labeling experiments. First, a $^{12}\text{C}/^{13}\text{C}$ switching experiment confirms acetaldehyde as the primary carbon source for the produced ethene. As shown in Fig. S8, co-feeding ^{13}C -methanol with natural abundance acetaldehyde (4:1) results in approximately 85.5% of the ethene molecules containing only ^{12}C atoms. Next, we conducted an H/D switching experiment using a mixture of CD_3OH and CH_3CHO (4:1) (Fig. 2b). The observed presence of monodeuterated ethene (54.0%) confirms the occurrence of hydrogen transfer between the reactants. As illustrated in Fig. 2c, a deuterium atom (D) from CD_3OH migrates to the carbonyl carbon of acetaldehyde, while the hydroxyl hydrogen from CD_3OH migrates to the carbonyl oxygen. This leads to the formation of monodeuterated ethanol (CH_3CHDOH), which subsequently dehydrates over zeolites to yield monodeuterated ethene. The significant amount of monodeuterated ethene provides strong evidence for the hydrogen transfer process. Some non-deuterated ethene is also observed, potentially arising from the Cannizzaro reaction of acetaldehyde. Additionally, a

minor fraction of multiply deuterated ethene is detected, likely originating from methanol conversion via the HP pathway at higher temperatures.

Identification of surface species during the MPV reduction over ZSM-5

^{13}C solid-state NMR spectroscopy was employed to investigate the transformation of surface species on zeolites during the MPV reaction between natural abundance methanol and ^{13}C -labeled acetaldehyde (left side, Fig. 3a) or ^{13}C -labeled methanol and natural abundance acetaldehyde (right side, Fig. 3a) at various temperatures (293 K, 373 K, 423 K, 473 K, 523 K and 573 K). This approach allows us to track the conversion of both reactants within the co-reaction system. Additionally, two-dimensional (2D) NMR experiments, including ^{13}C - ^{13}C proton-driven spin-diffusion (PDS) and ^1H - ^{13}C heteronuclear correlation (HETCOR) (Fig. 3b and Fig. S9), were used to identify the adsorbed species formed during the co-reaction (Fig. 3c).

The presence of methanol significantly impacts acetaldehyde reactivity even at low temperatures. At 293 K and 373 K, a series of consecutive dehydration reactions between acetaldehyde and methanol leads to the formation of 1,1-dimethoxyethane ($\text{CH}_3\text{OCH}(\text{CH}_3)\text{OCH}_3$, DMOE), as evidenced by specific ^{13}C NMR signals at 103, 54, and 19 ppm, along with corresponding ^1H signals at 4.9, 3.5, and 1.9 ppm (Fig. 3c). GC-MS analysis of the extract further corroborates this assignment (Fig. S10). The formation of DMOE is attributed to the strong polarity of methanol and its overloading in the co-reaction, leading to robust adsorption onto the BAS of zeolites. This facilitates methanol's interaction and reaction with acetaldehyde. Furthermore, the overloaded methanol promotes the formation of DMOE through multiple methylation reactions.

Acetaldehyde also undergoes aldol condensation over the BAS on zeolite⁵⁴, resulting in crotonaldehyde ($\text{CH}_3\text{CH}=\text{CHCHO}$) (Fig. 3c), as confirmed by the correlation peaks between the ^{13}C signals at 201, 133, and 19 ppm with the ^1H signals at 9.6, 6.6, and 1.9 ppm in the 2D NMR spectrum (Fig. 3b). This finding is further supported by GC-MS analysis of the extract (Fig. S10). However, the lower abundance of crotonaldehyde compared to DMOE suggests that a majority of the acetaldehyde reacts with methanol. As the temperature increases to 373 K, DMOE consumption accelerates, indicating its high reactivity (Fig. 2a, left). Surface ethoxy species (SES), observed at 72 ppm in the 1D ^{13}C CP/MAS NMR spectra (Figs. 3a) and 2D ^{13}C - ^{13}C PDS NMR spectrum (Fig. S9), reach a maximum concentration at 473 K and then decrease with further temperature rise. These species are a signature of the MPV reaction between acetaldehyde and methanol. Ethene can readily form from the decomposition of surface ethoxy species on zeolite surfaces⁵⁵.

At higher temperatures, acetaldehyde undergoes aromatization, evident by a broad ^{13}C NMR signal centered at ca. 135 ppm. These aromatic species, confirmed by GC-MS (Fig. S10), can act as HP precursors for light olefins production via reactions with ethene⁵⁰. Furthermore, aromatics can form through the intermediate methylcyclopentenone (MCPT), which is produced by the reaction of alkenes with oxygenates such as crotonaldehyde and acetaldehyde. The concentration of aromatics increases monotonically with temperature. Additionally, the presence of cyclopentenyl cations (CP^+) (240–260 ppm) above 473 K suggests the participation of the HP mechanism in acetaldehyde conversion^{2,56}, which is corroborated by the gas-phase product distribution (Fig. 2a). Ethene dominates at lower temperatures due to the MPV reaction, while C_3+ hydrocarbons become prominent at higher temperatures due to the HP mechanism. Finally, surface acetate species (183 ppm in the ^{13}C NMR spectrum), formed from acetaldehyde oxidation, are observed. While proposed as a key intermediate in the methanol conversion¹⁴, its near-constant signal below 573 K suggests limited transformation under these conditions.

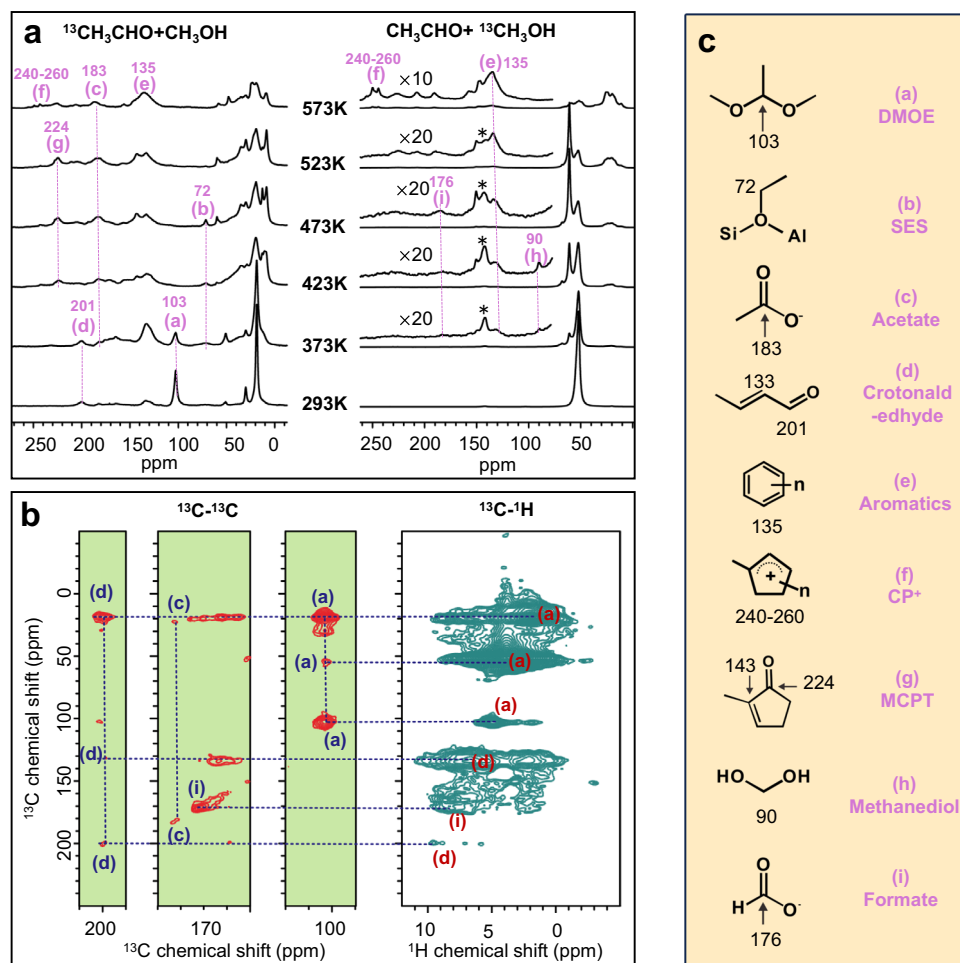


Fig. 3 | Identification of surface species in transformation of acetaldehyde with methanol by ^{13}C solid-state NMR spectroscopy. **a** 1D ^{13}C CP/MAS NMR spectra of the trapped species over ZSM-5 zeolites obtained from co-feeding natural abundance methanol and ^{13}C -acetaldehyde and co-feeding ^{13}C -methanol and natural abundance acetaldehyde at various temperatures for 8 s. Asterisks denote spinning

sidebands. **b** 2D ^{13}C - ^{13}C PDS and ^{13}C - ^1H HETCOR NMR spectra of the trapped species over ZSM-5 zeolites obtained from co-feeding ^{13}C -methanol and ^{13}C -acetaldehyde at 293 K for 8 s. **c** Identified surface species over ZSM-5 in MPV reaction.

^{13}C solid-state NMR spectroscopy was further employed to elucidate the transformation pathway of methanol during the MPV reaction. Co-reactions were conducted using ^{13}C -labeled methanol and natural abundance acetaldehyde, and the resulting surface species were monitored at various temperatures (Fig. 3a, right). At room temperature (293 K), the ^{13}C NMR spectrum reveals the presence of adsorbed methanol (45–53 ppm), consistent with its strong interaction with the BAS on zeolites. Additionally, a signal at 54 ppm indicates the formation of DMOE, mirroring the observation made with ^{13}C -labeled acetaldehyde (Fig. 3a, left). As the temperature increases to 373 K, a new ^{13}C signal emerges at 90 ppm, which is assigned to methanediol ($\text{HO}-\text{CH}_2-\text{OH}$) based on previous reports¹⁴. The appearance of methanediol suggests that methanol undergoes dehydrogenation via Brønsted acid-mediated hydrogen transfer reactions. This finding aligns with the concurrent formation of SES observed during acetaldehyde hydrogenation at this temperature (Fig. 2a, left), further supporting the MPV mechanism. Furthermore, methanediol can transform into surface formate species (176 ppm), which contributes to the formation of methyl formate observed in the effluent product (Fig. 2a). At even higher temperatures, methanol participates in the HP reactions. Similar to the observations with ^{13}C -labeled acetaldehyde, aromatics and cyclopentenyl cations are detected by ^{13}C NMR, signifying methanol involvement in secondary reactions that lead to the production of heavier hydrocarbon products.

Relationship between surface species and effluent products during the MPV reduction

The time-dependent transformation of acetaldehyde during methanol conversion over ZSM-5 at 523 K was investigated using ^{13}C NMR experiments to analyze surface species and GC to analyze effluent products (Fig. 4). As shown in Fig. 4a and Fig. S11, the ^{13}C NMR spectrum reveals the presence of SES even after a very short reaction time of 8 s. This indicates the immediate operation of the MPV reaction during the initial stages (Fig. 4c). Consequently, ethene dominates the effluent products at this point, with an E/P ratio of 2.82 (Fig. 4b). This dominance can be attributed to the decomposition of SES formed through the MPV reaction. Although some aromatics are also detected by ^{13}C NMR at this early stage, their contribution to the product stream seems limited, as evidenced by the minimal formation of C_{3+} hydrocarbons in the GC analysis. However, as the reaction progresses beyond 16 s, the ^{13}C NMR signal for SES rapidly diminishes, suggesting a weakening of the MPV reaction (Fig. 4a and Fig. S11). Conversely, the HP route appears to take precedence, although its overall reactivity remains limited at 523 K (Fig. 4c). This shift in reaction pathways is reflected in the product distribution (Fig. 4b). The selectivity of C_{3+} products, particularly propene, increases, leading to a decrease in the E/P ratio below 1. This observed trend aligns with typical methanol reaction over zeolites, where ethene initially dominates due to a more direct route, followed by a decrease as the HP pathway becomes more

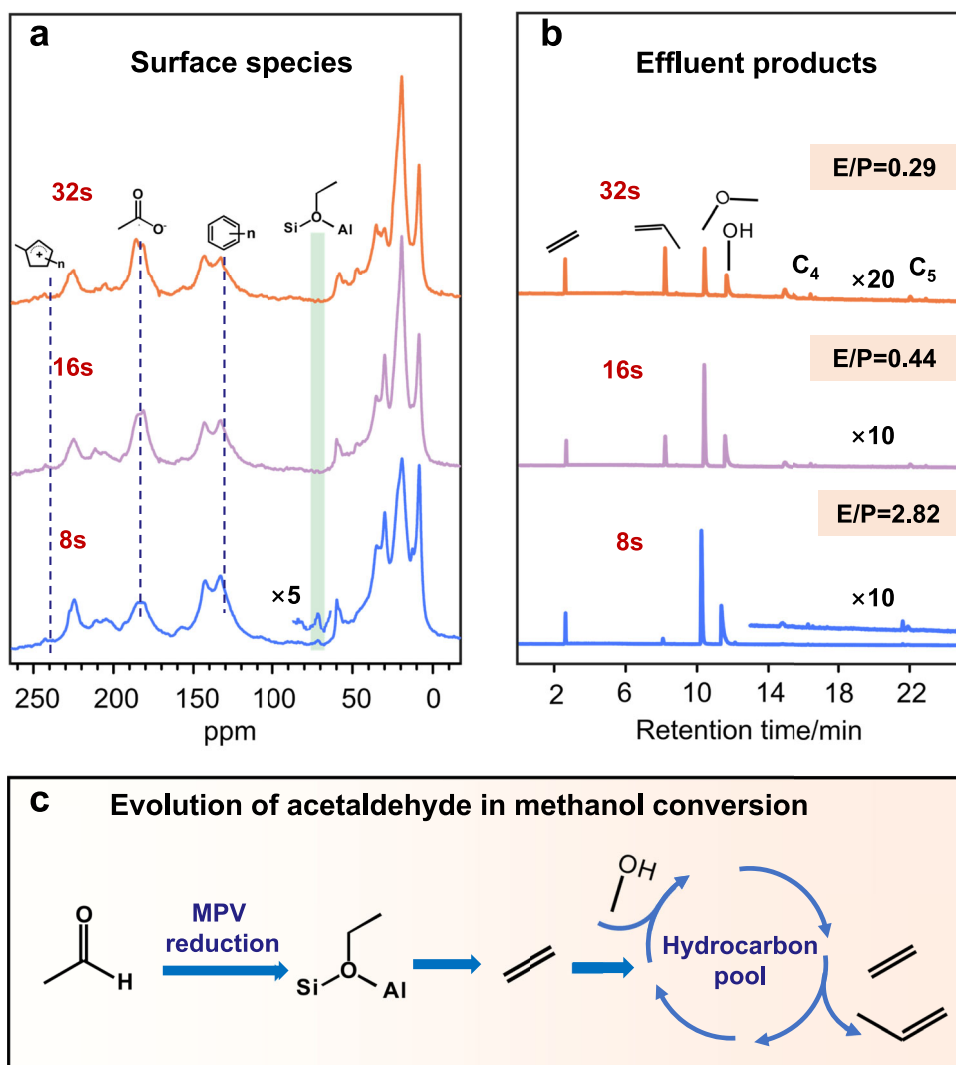


Fig. 4 | Time-dependent evolution of acetaldehyde in methanol conversion. **a** ^{13}C CP/MAS NMR spectra of the trapped species over ZSM-5 zeolites obtained from co-feeding natural abundance methanol and ^{13}C -acetaldehyde (v/v, 4/1) at 523 K for variable time. **b** GC analysis of effluent products from the co-reaction

between methanol and acetaldehyde at the same condition described in Fig. 3a. E/P indicates ethene to propene ratio. **c** Illustration of the evolution of acetaldehyde in MTH reaction.

favorable⁵⁷. These results suggest that the initial prevalence of ethene during the induction period in methanol conversion could be attributed to the early formation of acetaldehyde, which readily participates in the MPV reaction.

The MPV reaction between acetaldehyde and methanol over zeolites likely involves two main pathways, influenced by the adsorption behavior of reactants on active sites. ^{13}C NMR analysis suggests that methanol molecules and dimethyl ether dominate zeolite adsorption during the co-reaction (Fig. 3a). In-situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) experiments were performed to track the evolution of surface species on ZSM-5 zeolite during the co-reaction of methanol with acetaldehyde at 523 K (Fig. S12). This experiment reveals that methanol or dimethyl ether efficiently occupies the BAS. This occupation provides a sufficient supply of hydrogen atoms for the hydrogenation of acetaldehyde, thereby promoting the MPV reaction. This dominance is further supported by the near-first order reaction dependence of ethene formation on acetaldehyde concentration under high methanol atmospheres (Fig. S13). Essentially, most BAS become occupied by methanol or its derivatives (e.g., DME and surface methoxy species), limiting acetaldehyde adsorption.

Reaction mechanism of BAS catalyzed MPV reduction

Based on experimental observations and Density functional theory (DFT) calculations of activation and reaction energies for each elementary step (Figure S14), we propose a dual-pathway MPV mechanism (Fig. 5). In the pathway (a), methanol preferentially adsorbs on the BAS, forming protonated methanol (I) and surface methoxy species (II). Both intermediates are electrophiles that can interact with the oxygen atom of acetaldehyde. Subsequent dehydration between protonated methanol and acetaldehyde leads to the formation of the 1-methoxoethyl cation (III) ($\Delta E_{\text{act}} = 24.96$ kcal/mol). Similarly, methylation of acetaldehyde by surface methoxy species also generates species (III) ($\Delta E_{\text{act}} = 13.87$ kcal/mol). This carbocation, with its electron-deficient structure, captures a hydride atom from methanol's methyl group, resulting in the formation of methyl ethyl ether (IV) and formaldehyde. This hydride transfer step, although endothermic ($\Delta E_{\text{rea}} = 3.72$ kcal/mol), exhibits a moderate activation energy ($\Delta E_{\text{act}} = 25.97$ kcal/mol). Methyl ethyl ether readily transforms to SES and then ethene. Additionally, carbocation (III) can react with another methanol molecule to form DMOE (V) ($\Delta E_{\text{act}} = 0.75$ kcal/mol), detected by ^{13}C NMR in the co-reaction of acetaldehyde and methanol at low reaction temperatures (Fig. 3a). This species can reversibly transform to

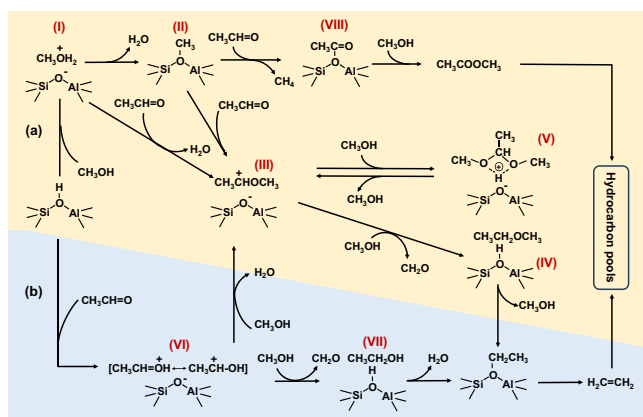


Fig. 5 | Proposed reaction routes for the transformation of acetaldehyde in the presence of methanol over BAS within zeolites. MPV reactions in methanol conversion over ZSM-5 based on the preferential adsorption of methanol (a) and acetaldehyde (b) on zeolite.

carbenium (III) and subsequently undergo the MPV reaction to yield ethene via intermediate (IV).

The alternative pathway (b) involves acetaldehyde adsorption on the BAS, forming a protonated acetaldehyde that resonates with carbenium (VI). This carbenium captures a hydride from methanol's methyl group, leading to the formation of ethanol (VII) and formaldehyde. Although endothermic ($\Delta E_{\text{rea}} = 15.47$ kcal/mol), this hydride transfer step possesses a lower activation energy ($\Delta E_{\text{act}} = 21.49$ kcal/mol). The formed ethanol can subsequently transform into ethene via SES. While not the primary pathway for the hydride transfer reaction between acetaldehyde and methanol, this route (pathway (b)) contributes to the co-reaction. Furthermore, the methylation of intermediate (III) with methanol can facilitate the generation of intermediate (VI) ($\Delta E_{\text{act}} = 29.37$ kcal/mol), thereby resulting in the MPV reaction in Pathway (a). In addition, the carbenium (VI) can initiate side reactions like aldol condensation and the Cannizzaro reaction.

Furthermore, surface methoxy species, due to their electron-deficient nature, can also accept a hydride from acetaldehyde, producing methane and acetate (VIII). Although it is an endothermic reaction ($\Delta E_{\text{rea}} = 12.16$ kcal/mol) with a relatively high activation energy ($\Delta E_{\text{act}} = 33.56$ kcal/mol), this hydride transfer pathway remains viable under the overall reaction conditions. The acetate observed in the ^{13}C MAS NMR spectra (Fig. 3) can further react with methanol to form methyl acetate. Both ethene and methyl acetate can then transform into HP species^{15,28}, initiating the complex methanol conversion process.

The entire reaction network for the MPV process indicates that hydride transfer between acetaldehyde and methanol, or related derivatives, occurs on the BAS of zeolites. This MPV reaction exhibits a comparatively low activation energy, suggesting its preferential role in the initial stages of the process. Ethene and formaldehyde are preferentially formed via MPV reduction before the generation of heavier hydrocarbons. This establishes a viable reaction pathway that connects the initial C-C bond species to the hydrocarbon pool species during methanol conversion. It is hypothesized that aldehydes/ketones with similar properties to acetaldehyde would likely undergo MPV reduction with methanol over zeolite catalysts. To expand this concept, we investigated the transformation of acetone in an overloaded methanol environment ($v_{\text{methanol}}/v_{\text{acetone}} = 4/1$) over ZSM-5 zeolite at various temperatures (423–573 K). Analysis of the effluent stream by GC reveals propene as the primary product with high selectivity (propene to ethene ratio, P/E = 7.56) in this co-reaction at 573 K (Fig. S15). This observation can be attributed to MPV reduction occurring between acetone and methanol, leading to the formation of

isopropanol or surface isopropoxy species. These surface intermediates subsequently decompose into propene. The presence of the crucial intermediate, the surface isopropoxy species, within the co-reaction is confirmed using ^{13}C MAS NMR spectroscopy (Fig. S16). The MPV reduction reactions between methanol and both acetaldehyde and acetone highlight the critical role of this pathway in transforming aldehydes and ketones. This mechanistic understanding underscores the pivotal role that MPV reduction plays within the oxygenate cycle during methanol conversion processes.

Discussion

This study unveils a previously unrecognized facet of methanol conversion over zeolites, demonstrating the ability of the BAS within ZSM-5 zeolites to catalyze an MPV-type reaction between acetaldehyde and methanol. The proposed mechanism involves the formation of carbocation intermediates via two distinct pathways: (1) methylation by surface methoxy species and (2) BAS-mediated protonation of acetaldehyde. Subsequent hydride transfer from methanol to these electron-deficient cations leads to the generation of ethanol or surface ethoxy species. This convertible precursor readily facilitates the formation of ethene, the primary alkene product observed in methanol conversion. Interestingly, elevated reaction temperatures favor an alternative pathway involving hydride transfer between methoxy species and acetaldehyde. This could potentially lead to the formation of methyl acetate, a crucial intermediate in the established hydrocarbon pool mechanism. The BAS-catalyzed MPV reaction demonstrates versatility towards aldehydes/ketones, exemplified by the selective production of propene from acetone. Aldehydes and ketones are common reactive intermediates during methanol conversion over zeolites, especially during the induction period. Their high reactivity makes them susceptible to undesirable side reactions, including coke formation, which is a major challenge in MTH processes. The MPV reduction provides an alternative reaction pathway, converting these carbonyl compounds into lower alkenes. This process effectively shortens the induction period and delays the onset of coke formation.

The identification of this unconventional MPV-type pathway not only broadens the understanding of oxygenate transformation during methanol conversion but also suggests potential avenues for the design of more efficient and selective catalysts. By strategically optimizing the distribution and accessibility of BAS over the zeolite catalysts, the transformation of aldehydes and ketones in the methanol conversion process can be effectively regulated. For instance, by creating BAS pairs or enriching the internal or external surfaces with BAS, the catalytic pathways leading to the formation and subsequent conversion of these intermediate carbonyl species can be precisely controlled. This tailored distribution allows for precise control over reaction efficiency and product selectivity, ultimately leading to improved overall performance in the methanol conversion process.

Methods

Catalyst preparation and characterization

Proton-type ZSM-5 zeolites (Si/Al = 11.5, Zeolyst) were utilized as catalysts for all the reactions unless otherwise indicated. A control ZSM-5 sample enriched with Lewis acid sites was prepared by calcination in air at 973 K for 6 h. The sample was packed into a shallow bed configuration on a quartz crucible. The temperature was increased gradually from room temperature to the target temperature at a rate of $1^\circ\text{C}/\text{min}$. The structure and crystalline nature of ZSM-5 zeolites were examined by a PANalytical X'Pert Powder X-ray diffraction diffractometer with a Cu K α radiation using a step of 0.02° at a respective voltage of 40 kV and a current of 40 mA. The scanning range was 5° to 40° .

The acid concentration of catalysts was determined by IR of pyridine adsorption experiments. The measurements were performed on a Bruker Tensor 27 spectrometer. The catalysts were first activated at

673 K under high vacuum ($<10^{-5}$ Pa) for 12 h. After cooling down to 298 K, a background spectrum was collected. Physisorbed pyridine was removed by evacuation at 473 K for 1 h and then held for 2 h to reach equilibrium. The IR spectra of pyridine-adsorbed samples were measured at 293 K after evacuation.

Catalytic evaluation

All catalytic reactions were carried out in a pulse quench reactor with an inner diameter of 6 mm. Before the reaction, the ZSM-5 powder was pressed into pellets between 40–60 mesh. 0.1 g pellets were loaded into the reactor and activated at 400 °C in flowing high purity He (99.999%) for 1 hour and then temperature programmed cooled down to reaction temperature. The total gas flow through the reactor was stabilized at 200 sccm. Then the reaction was thermally quenched by pulsing liquid nitrogen onto the catalyst bed, and the temperature was cooled down from reaction temperature to ambient temperature within a very short period (0.5 s), which was achieved using high-speed valves controlled by a GC computer. In each case, a certain amount of reactants was pulsed into the reactor containing 0.1 g ZSM-5 zeolites and reacted for target time at the reaction temperature. For the co-reaction of methanol and acetaldehyde, methanol and acetaldehyde with different volume ratio were co-fed onto the catalyst and allowed to react for target time at different reaction temperatures. The total gas flow through the reactor was stabilized at 200 sccm. The real-time analysis of effluent products was analyzed by GC-MS online after the pulse quench reaction. The retained products were also analyzed by GC-MS. Namely, the zeolites with retained species were dissolved in 1 ml 20 wt% HF solution and then extracted with 1 ml CH_2Cl_2 . After uniformly shaking and mixing, standing for enough time, the bottom layer containing the organic phase of the extracted solution was separated and analyzed by GC-MS. The catalyst with retained products was directly analyzed by solid-state NMR spectroscopy. After the reaction was quenched, the reactor containing the catalyst was cooled down by plenty of liquid nitrogen and sealed with valves. The sealed reactor was then transferred to a glove box filled with high-purity N_2 (99.999%) and the catalyst was packed into 4 mm NMR rotor for NMR measurements. The effluent products at the moment of quenching were analyzed using GC-MS to establish a correlation between the effluent products and the surface-adsorbed species on zeolites.

The effluent products were analyzed by GC-MS (Shimadzu GCMS QP-2010) equipped with RT-Q-BOND capillary column (30 m, 0.32 mm i.d., 10.0 μm film thickness) or Petrocol DH capillary column (100 m, 0.25 mm i.d., 0.5 μm film thickness). The temperature programming of RT-Q-BOND capillary column started at 40 °C (maintained for 1 min), followed by a rate of 5 °C per min to a 140 °C and a rate of 10 °C per min to a final temperature of 230 °C (maintained for 5 min). The temperature programming of Petrocol DH capillary column started at 40 °C (maintained for 2 min), followed by a rate of 1 °C per min to a 50 °C (maintained for 5 min) and a rate of 15 °C per min to a final temperature of 230 °C. When checking the effluent products, the line from the reactor outlet to the GC was maintained at 100 °C to avoid any condensation of heavy hydrocarbons. The retained species were also analyzed by GC MS (Shimadzu GCMS QP-2010) equipped with Petrocol DH capillary column (100 m, 0.25 mm i.d., 0.5 μm film thickness). The following temperature programming was applied: an initial temperature of 40 °C was maintained for 20 min, followed by a rate of 10 °C per min to 220 °C (maintained for 15 min) and a rate of 15 °C per min to 250 °C (maintained for 5 min).

Isotopic labeling experiments

In $^{13}\text{C}/^{12}\text{C}$ or H/D isotope switching experiments, the ^{13}C -labeled methanol (^{13}C 99 %) and acetaldehyde in natural abundance were co-reacted on H-ZSM-5 zeolites at 523 K for 8 s. Methanol- d_3 (D, 99 %) and acetaldehyde in natural abundance was co-reacted on ZSM-5 at 523 K for 8 s. The isotopic distribution was determined MS analysis.

Solid-state NMR measurements

The ^{13}C , ^{27}Al , ^{29}Si solid-state NMR experiments were carried out at 9.4 T on a Bruker Avance III 400 spectrometer, equipped with a 4 mm probe, with resonance frequencies of 399.33 and 100.42 MHz for ^1H and ^{13}C , respectively. The magic angle spinning (MAS) rate was set to 10 kHz. Adamantane was used as an external chemical shift reference for ^{13}C by setting the CH_2 resonance to 38.5 ppm, and ^{27}Al NMR spectra were referenced with respect to aqueous $\text{Al}(\text{NO}_3)_3$ at 0 ppm. For the $^1\text{H} \rightarrow ^{13}\text{C}$ CP/MAS NMR experiments, the Hartmann-Hahn condition was achieved using hexamethylbenzene (HMB), with a contact time of 1.5 ms and a repetition time of 2 s. 2D $^{13}\text{C}\{-^1\text{H}\}$ HETCOR were acquired with 64 scans for each of 128 experiments with t_1 increments of 55.56 μs . For ^{27}Al single-pulse NMR experiments, the ^{27}Al spectra were accumulated for 1024 scans with $\pi/12$ flip angle, and 2 s repetition time. Single-pulse ^{29}Si MAS NMR spectra were recorded using a $\pi/2$ pulse of 5 μs , a recycle delay of 60 s, and a spinning rate of 10 kHz. The ^{29}Si chemical shifts were referenced to kaolinite (-91.5 ppm). The 2D $^{13}\text{C}\text{-}^{13}\text{C}$ PDS (proton-drive-spin-diffusion) spectra were recorded at 10 kHz MAS, a 70 % ramp for the ^1H CP pulse with a 2.5 ms CP contact time, 2 s recycle delay, 12 ms and 8 ms acquisition time for direct and indirect ^{13}C dimensions, respectively, and 300 ms for the proton-driven spin-diffusion mixing time.

Calculation method

The 88 T cluster model, depicted in Fig. S17, was utilized to represent the ZSM-5 zeolite, based on its crystallographic structural data available at (<http://www.iza-structure.org/databases/>). This model captures the full range of the double 10-membered ring (10-MR) intersection pores characteristic of ZSM-5. An aluminum (Al) atom was specifically positioned at the T12 site, while a proton was anchored at the O24 location. Furthermore, the terminal silicon-hydrogen (Si-H) bond was precisely measured at 1.47 Å, oriented along the direction of the corresponding Si-O bond.

In the current research, the 30 T active site atoms along with the adsorbed hydrocarbon complex were categorized as the high-level layer (Fig. S17), with the remaining framework serving as the low-level layer. To maintain the integrity of the modeled zeolite, we performed targeted optimizations, permitting the relaxation of atoms in the high-level layer while keeping the others fixed in their crystallographic positions. The Quadratic Synchronous Transit (QST3) method was applied to locate all transition state (TS) structures. By utilizing the imaginary vibrational mode of the optimized TS, we carefully modified the positions of the vibrational atoms along the computed reaction coordinate in both directions towards the reactant and product, further refining these structures to achieve their minimum energy states.

For optimizing both adsorption and transition states, we adopted a hybrid theoretical method called ONIOM, combining $\omega\text{B97XD}/6\text{-}31\text{G(d,p)}$ for the high-level layer and AM1 for the low-level layer. The ωB97XD method is well-known for effectively addressing both short- and long-range dispersion interactions and has been widely employed in exploring reaction mechanisms in zeolites^{58,59}. To address the potential underestimation of low-level interaction energies by AM1, all reported energies were recalculated at the $\omega\text{B97XD}/6\text{-}31\text{G(d,p)}$ level, based on the optimized structures derived from the ONIOM methodology.

Data availability

All data supporting the findings of this study are available within the article and Supplementary Information files. All data are available from the corresponding author upon request. Source data are provided with this paper.

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Author contributions

W.J.C., C.W., and M.H. conducted the catalysts preparation, characterization, and catalytic testings; Y.Y.C. performed the DFT calculations; W.J.C. and Q.W. carried out the solid-state NMR experiments; W.J.C., C.W., and J.X. wrote the manuscript; and C.W., J.X., and F.D. designed and directed the project. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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